

Report
Of
SIEMENS DATA SCIENCE MASTER VIRTUAL INTERNSHIP
(AICTE – EduSkills Virtual Internship Program)

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE (Cyber Security)



By
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Batch: 2023-2027

2023-2027

CANDIDATE'S DECLARATION

I, AYUSH MITTAL, do hereby declare that the internship report titled “Siemens Data Science Master Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Cyber Security). All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: AYUSH MITTAL

Roll No.: CS-23411189

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the AICTE – EduSkills Virtual Internship Program on Siemens Data Science Master. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank EduSkills Foundation, the All India Council for Technical Education (AICTE), and Siemens for providing this valuable virtual internship opportunity, and my institute, IILM University, Greater Noida, for its continuous encouragement and support throughout the program.

I express my gratitude to my parents for their blessings.

Signature

Student Name: AYUSH MITTAL

Roll No.: CS-23411189

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3. Internship Completion Certificate



Figure 3.1: Certificate of Virtual Internship issued by AICTE – EduSkills – Siemens

4. Project Description

4.1 Introduction

This report presents the work carried out during the 8-week “Siemens Data Science Master” Virtual Internship, conducted under the AICTE – EduSkills National Internship Program from April 2026 to June 2026. The internship was undertaken by Ayush Mittal, a B.Tech (Computer Science and Engineering – Cyber Security) student of IILM University, Greater Noida, under AICTE Student ID STU664a2059862571716133977.

The program was designed to bridge the gap between academic learning and industry expectations by providing structured, week-wise, hands-on training in data science and machine learning using the RapidMiner platform. The internship combined guided learning modules, weekly assessments, project documentation, and a final assessment test to evaluate practical proficiency.

4.2 Organization Profile

The internship was jointly conducted by the following organizations:

- AICTE (All India Council for Technical Education) – the statutory body of the Ministry of Education, Government of India, responsible for planning and coordinated development of technical education, and the parent body of the National Internship Portal.
- EduSkills Foundation – an ISO 9001:2015 and ISO 20000-1:2018 certified, ‘Great Place to Work’ certified organization that partners with industry and academia to deliver skill-based virtual internship programs under the theme “Nation Building Through Skills.”
- Siemens (Siemens Industry Software (India) Pvt. Ltd.) – the industry partner that sponsored and supported the Data Science Master internship domain, providing the RapidMiner platform and curriculum content.

Together, these organizations administer the AICTE – EduSkills Virtual Internship Program, which enables students across India to gain industry-relevant, credentialed exposure to emerging technologies without requiring physical relocation.

4.3 Problem Statement

Modern organizations generate large volumes of data but often lack professionals who can translate that data into actionable business decisions using structured data science methods. Academic curricula frequently emphasize theoretical concepts without providing sufficient hands-on exposure to enterprise-grade analytics platforms. This

creates a skill gap between what industry expects from a data science professional and what a typical graduate is trained to do.

The Siemens Data Science Master internship was designed to address this problem by training the intern to identify real-world business problems suitable for AI/ML, and to solve them end-to-end — from data access and preparation to model building, deployment, and platform administration — using the industry-grade RapidMiner AI Hub environment.

4.4 Project Objectives

The key objectives of the internship were:

- To learn how to identify business problems that are suitable for AI and machine learning solutions.
- To gain practical skills in accessing, loading, transforming, and calculating data.
- To build foundational and advanced machine learning skills, including model building and evaluation.
- To master advanced data preparation and automation techniques.
- To learn how to deploy and monitor machine learning models in a production-like setting.
- To gain exposure to configuring and administering the RapidMiner AI Hub at an enterprise level.
- To complete all weekly assessments, project documentation, and the final assessment test in order to earn the Virtual Internship Certificate.

4.5 Scope of the Project

The scope of this internship covers the complete data science lifecycle as implemented on the RapidMiner platform – from problem identification and data engineering to machine learning model development, deployment, monitoring, and platform administration. The concepts and techniques learned are applicable across multiple domains such as manufacturing, finance, healthcare, and retail, wherever data-driven decision-making is required.

The scope of this report is limited to the coursework, weekly modules, assessments, and project work completed during the 8-week virtual internship period (April 2026 – June 2026), and does not extend to any live client or production deployment, since the internship was conducted in a guided, sandboxed virtual-lab environment.

4.6 Technologies and Tools Used

- RapidMiner AI Hub / RapidMiner Studio – for data access, preparation, model building, deployment, and platform administration.
- Statistical and Machine Learning algorithms – including regression, classification, and clustering techniques.
- Data visualization and dashboarding tools – for presenting analytical results.
- AICTE – EduSkills online learning portal (LMS) – for structured week-wise modules, assessments, and the final assessment test.
- Cloud-based virtual lab environment – for hands-on practice without local infrastructure requirements.

4.7 System Architecture

The data science workflow followed during the internship can be represented as a pipeline of sequential stages, moving from raw data sources to a fully deployed and monitored model, administered through the RapidMiner AI Hub. The stages of this architecture are summarized below.

Stage	Component	Description
1	Data Sources	Enterprise databases, spreadsheets, flat files and cloud repositories used as raw input.
2	Data Access & Ingestion	Connecting to and loading data into the RapidMiner AI Hub environment.
3	Data Transformation & Cleaning	Handling missing values, formatting, and calculating derived attributes.
4	Feature Engineering	Preparing and selecting relevant attributes for model building.
5	Model Building	Applying regression, classification and clustering algorithms.
6	Model Validation	Evaluating model performance using cross-validation and accuracy metrics.
7	Deployment & Monitoring	Publishing models as services and monitoring them in production.
8	Platform Administration	Managing users, roles, and enterprise operations on RapidMiner AI Hub.

Table 4.1: System Architecture Stages

4.8 Methodology

The internship followed a structured, week-wise methodology as prescribed in the official offer letter (Ref No. C16Y2026M081417437) issued by EduSkills. Each week focused on a specific module, progressing from foundational to advanced and enterprise-level competencies, as summarized in the table below.

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.

Table 4.2: Week-wise Training Methodology

In the final week, a Final Assessment Test was conducted, and the overall grade was computed based on performance in weekly assessments, project submissions, and the final test.

4.9 Expected Outcomes

Upon completion of the internship, the following outcomes were achieved:

- Ability to identify business problems suitable for AI/ML-based solutions.
- Practical proficiency in data engineering – accessing, loading, transforming, and calculating data.
- Hands-on skills in building, validating, and deploying machine learning models.
- Working knowledge of enterprise platform administration on RapidMiner AI Hub.
- Successful completion of the 8-week program with an Outstanding (Grade ‘O’) result, Certificate ID: 4ba590e3058ccd151850.
- Enhanced employability and confidence in applying data science techniques to real-world problems.

4.10 Certificates of Completion and Other Communication Proof

The Internship Completion Certificate issued by AICTE – EduSkills – Siemens is attached as Figure 3.1 in Chapter 3 of this report. The official Internship Offer Letter (Ref No. C16Y2026M081417437), which served as the communication proof for enrolment into the program, is attached below.

Ref No: C16Y2026M081417437

Internship Offer Letter

Student Name : AYUSH MITTAL
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering (Cyber Security))
Institute Name : IILM University, Greater Noida
Internship Domain : Siemens Data Science Master
Internship Duration : April 2026 - June 2026
AICTE Student ID : STU664a2059862571716133977

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

Figure 4.1: Internship Offer Letter – Page 1

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



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Figure 4.2: Internship Offer Letter – Page 2

As additional proof of successful completion, email confirmations received for the AICTE–EduSkills Internship Certificate and the RapidMiner Academy Professional Certification are attached below.

AICTE–EduSkills Internship 2024 Certificate download link External Inbox x

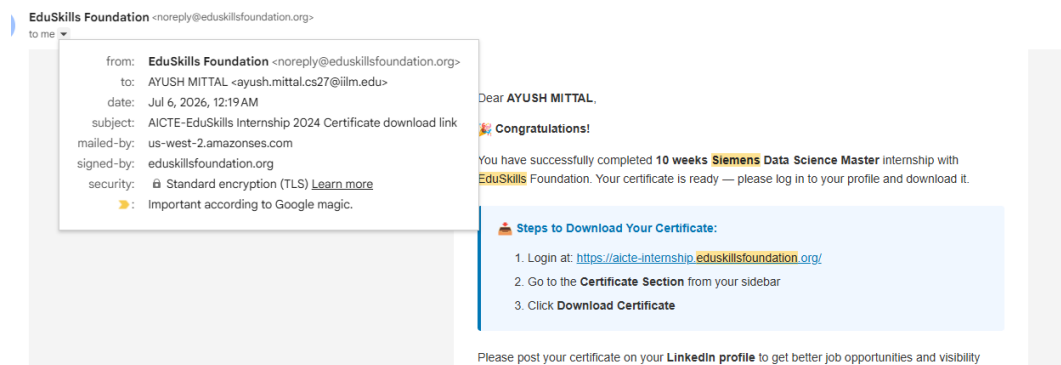


Figure 4.3: Email Confirmation – AICTE-EduSkills Internship Certificate Download Link

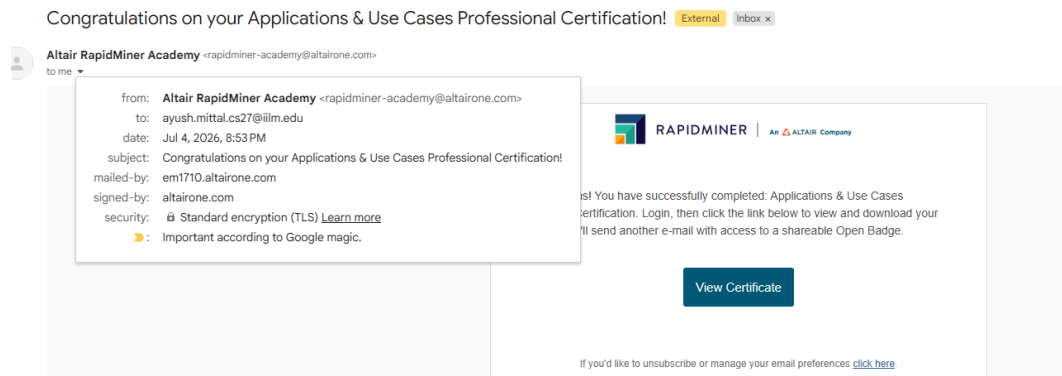


Figure 4.4: Email Confirmation – RapidMiner Academy Applications & Use Cases Professional Certification

5. Bibliography/References

1. AICTE – National Internship Portal, All India Council for Technical Education, Ministry of Education, Government of India. Available at: <https://internship.aicte-india.org>
2. EduSkills Foundation – “Nation Building Through Skills.” Available at: <https://eduskillsfoundation.org>
3. Siemens Digital Industries Software. Available at: <https://plm.sw.siemens.com>
4. RapidMiner Official Documentation. Available at: <https://docs.rapidminer.com>
5. Internship Offer Letter, Ref No. C16Y2026M081417437, issued by EduSkills to Ayush Mittal, 2026.
6. Certificate of Virtual Internship, Certificate ID: 4ba590e3058ccd151850, Student ID: STU664a2059862571716133977, issued by AICTE – EduSkills – Siemens, 2026.
7. AICTE–EduSkills Internship Certificate Download Link email, received from EduSkills Foundation (noreply@eduskillsfoundation.org), dated 6 July 2026.
8. Applications & Use Cases Professional Certification confirmation email, received from Altair RapidMiner Academy (rapidminer-academy@altairone.com), dated 4 July 2026.